Low-loss contacts on textured substrates for inverted perovskite solar cells

https://doi.org/10.1038/s41586-023-06745-7

Received: 28 June 2023

Accepted: 12 October 2023

Published online: 23 October 2023

Check for updates

So Min Park[1,2,10], Mingyang Wei[3,10], Nikolaos Lempesis[4,10], Wenjin Yu[5], Tareq Hossain[6], Lorenzo Agosta[4], Virginia Carnevali[4], Harindi R. Atapattu[6], Peter Serles[7], Felix T. Eickemeyer[3], Heejong Shin[1], Maral Vafaie[2], Deokjae Choi[1], Kasra Darabi[8], Eui Dae Jung[2], Yi Yang[1], Da Bin Kim[2], Shaik M. Zakeeruddin[3], Bin Chen[1], Aram Amassian[8], Tobin Filleter[7], Mercouri G. Kanatzidis[1], Kenneth R. Graham[6], Lixin Xiao[5], Ursula Rothlisberger[4], Michael Grätzel[3]✉ & Edward H. Sargent[1,2,9]✉

Inverted perovskite solar cells (PSCs) promise enhanced operating stability compared to their normal-structure counterparts[1–3]. To improve efficiency further, it is crucial to combine effective light management with low interfacial losses[4,5]. Here we develop a conformal self-assembled monolayer (SAM) as the hole-selective contact on light-managing textured substrates. Molecular dynamics simulations indicate that cluster formation during phosphonic acid adsorption leads to incomplete SAM coverage. We devise a co-adsorbent strategy that disassembles high-order clusters, thus homogenizing the distribution of phosphonic acid molecules, and thereby minimizing interfacial recombination and improving electronic structures. We report a laboratory-measured power conversion efficiency (PCE) of 25.3% and a certified quasi-steady-state PCE of 24.8% for inverted PSCs, with a photocurrent approaching 95% of the Shockley–Queisser maximum. An encapsulated device having a PCE of 24.6% at room temperature retains 95% of its peak performance when stressed at 65 °C and 50% relative humidity following more than 1,000 h of maximum power point tracking under 1 sun illumination. This represents one of the most stable PSCs subjected to accelerated ageing: achieved with a PCE surpassing 24%. The engineering of phosphonic acid adsorption on textured substrates offers a promising avenue for efficient and stable PSCs. It is also anticipated to benefit other optoelectronic devices that require light management.

Perovskite solar cells (PSCs) have recently attained a certified efficiency of 26.1% (ref. 6); however, the very highest power conversion efficiency (PCE) devices have yet to meet operating stability under accelerated ageing tests[5]. The limited device stability is attributed to the presence of mobile and hygroscopic p-type dopants in hole-transporting layers, which undermine moisture and thermal stability[7]. Inverted PSCs present a solution by utilizing undoped hole-selective contacts[8]. Recent studies have shown PCEs surpassing 25% for inverted PSCs[2]. However, when adhering to the stringent quasi-steady-state (QSS) protocol, their certified efficiency (approximately 24%) requires further improvement[3,9].

Efforts to improve inverted PSC efficiency have focused on interface passivation[2,9,10]. This approach suppresses non-radiative recombination and thus improves fill factors and photovoltages[11]. Nevertheless, higher photocurrents are needed for efficiency gains. In normal-structure PSCs, this has been realized using textured fluorine-doped tin oxides (FTO) as the transparent conductive oxides (TCO)[4,5]. The pyramidal grains on FTO minimize reflection losses and extend the length of the average optical path[4]. By contrast, inverted PSCs, often built on smooth indium tin oxides (ITOs)[9,10], face substantial optical losses due to the lack of light management.

The discrepancy is ascribed to differences in transporting materials and deposition techniques. In normal-structure PSCs, inorganic hole-blocking layers are conformally deposited on textured substrates using spray pyrolysis and chemical bath deposition[4,5]. However, inverted PSCs typically use ultrathin (less than 5 nm) organic hole-selective contacts on substrates[2,3], which can be challenging to deposit uniformly using solution processing[12]. Inhomogeneity leads to energy losses and insufficient carrier extraction[13].

Self-assembled monolayers (SAMs), particularly those composed of phosphonic acid molecules with hole-selective tail groups, have shown promise in addressing this issue[14]. Phosphonic acids establish coordinative/covalent bonds with TCOs, enabling sufficient SAM coverage

[1]Department of Chemistry, Northwestern University, Evanston, IL, USA. [2]Department of Electrical and Computer Engineering, University of Toronto, Toronto, Ontario, Canada. [3]Laboratory of Photonics and Interfaces, Ecole Polytechnique Fédérale de Lausanne, Lausanne, Switzerland. [4]Laboratory of Computational Chemistry and Biochemistry, Ecole Polytechnique Fédérale de Lausanne, Lausanne, Switzerland. [5]State Key Laboratory for Artificial Microstructure and Mesoscopic Physics, Department of Physics, Peking University, Beijing, P. R. China. [6]Department of Chemistry, University of Kentucky, Lexington, KY, USA. [7]Department of Mechanical and Industrial Engineering, University of Toronto, Toronto, Ontario, Canada. [8]Department of Materials Science and Engineering, and Organic and Carbon Electronics Laboratories (ORaCEL), North Carolina State University, Raleigh, NC, USA. [9]Department of Electrical and Computer Engineering, Northwestern University, Evanston, IL, USA. [10]These authors contributed equally: So Min Park, Mingyang Wei, Nikolaos Lempesis. ✉e-mail: michael.graetzel@epfl.ch; ted.sargent@northwestern.edu

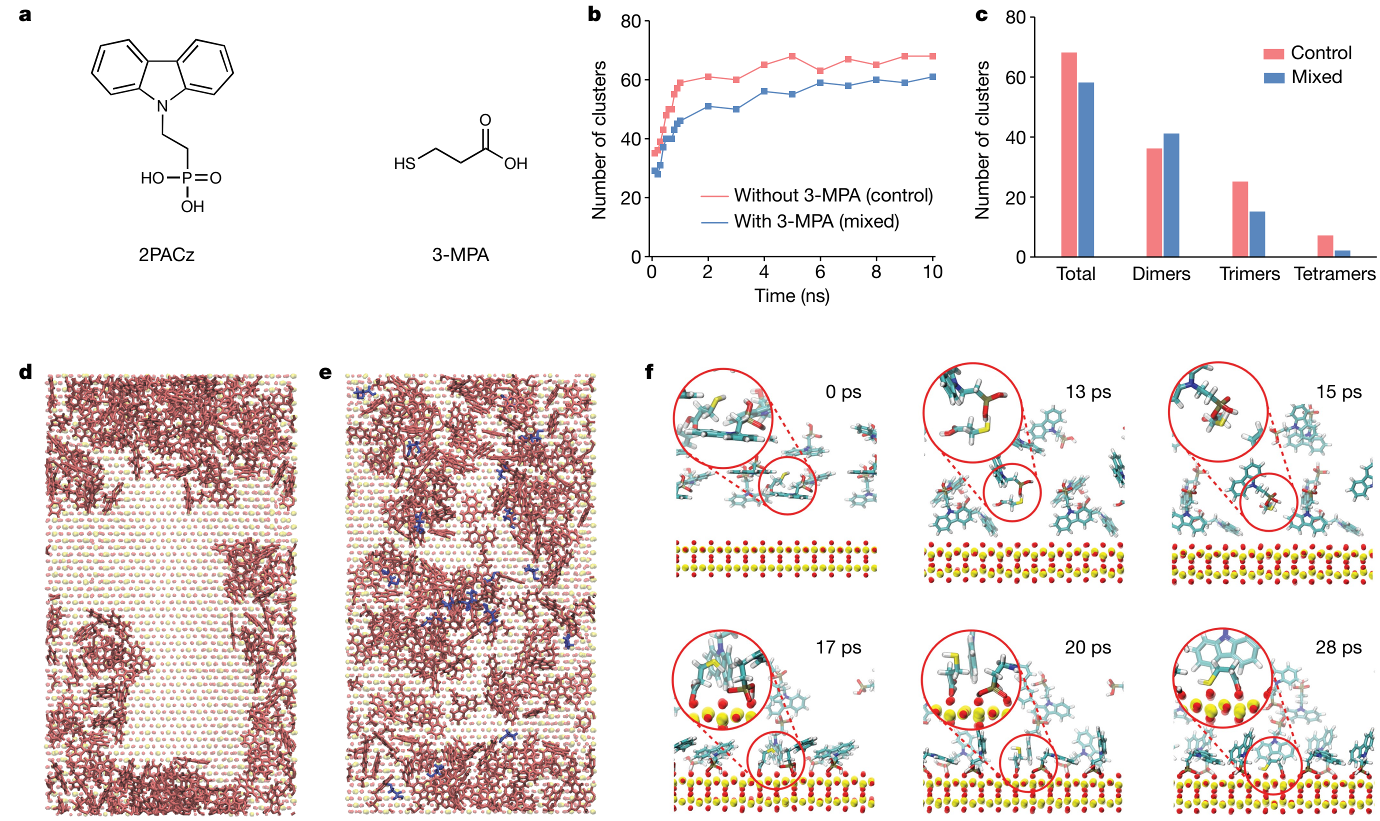

Fig. 1 | Molecular dynamics simulations of phosphonic acid adsorption with and without molecular additives. a, Chemical structures of the phosphonic acid 2PACz and the bifunctional compound 3-MPA. **b**, Total number of 2PACz clusters formed over a set period, in the absence (without, control) and presence (with, mixed) of 3-MPA. **c**, Types of 2PACz clusters formed at equilibrium. **d**,**e**, Top views of equilibrated molecular representations of control (**d**) and mixed (**e**) systems. 2PACz and 3-MPA (where applicable) are shown in pink and blue, respectively; Sn and O atoms, shown in the background, are depicted in yellow and red, respectively. **f**, Successive steps along an AIMD trajectory showcasing the role of 3-MPA as a co-adsorbent. Large, encircled areas are magnifications of small ones.

on rough surfaces[15]. This could provide a low-loss interface, uniting light management with interface passivation. Indeed, photocurrent improvements have been observed when replacing SAM-modified ITO with FTO substrates in inverted PSCs[16].

Despite their promise, achieving a high-density, closely packed SAM remains challenging, which leads to erratic interfacial properties[17]. For example, Liu and co-workers revealed that phosphonic acids failed to fully cover textured wafers, which affected the performance of tandem solar cells[18]. Even on flat substrates, the formation of SAMs tends to be non-uniform, a problem usually attributed to the limited solubility of phosphonic acids[19] and their insufficient chemical bond formation with metal oxides[20].

Several strategies have been proposed to ensure conformal SAM coverage, such as utilizing more reactive oxide surfaces (for example, nickel oxides) to promote hetero condensation[18], using thermal evaporation for SAM deposition[21] and designing highly soluble phosphonic acid molecules[19]. However, these methods can be laborious, requiring time-intensive molecule design, synthesis and vacuum-based layer deposition. Additionally, the use of highly reactive oxides could compromise device stability owing to redox reactions with ammonium halides[22].

Molecular dynamics simulations

We sought to deepen understanding of how phosphonic acids interact with textured TCOs. We began by looking at the case of 2-(9*H*-carbazol-9-yl)ethyl)phosphonic acid (2PACz), an organic molecule (Fig. 1a) capable of forming SAMs on TCOs[14], and investigated its interactions with FTO. 2PACz has found extensive applications as a hole-selective contact in PSCs in view of its excellent defect passivation and deep highest occupied molecular orbital level[21]. The surface properties of FTO were approximated using tin oxides (SnO_2), since commercial FTO has a fluorine-doping level under 0.5% (atomic percentage)[23]. Flat versus corrugated SnO_2 was analysed to understand the effects of surface morphology (Supplementary Fig. 1).

Molecular dynamics simulations suggest that 2PACz molecules can agglomerate, forming dimers, trimers and tetramers (Fig. 1b and Supplementary Fig. 2). Within the initial 1 nanosecond of simulations, 90% of the final number of clusters has been reached (Fig. 1b), and dimers were the most prevalent among those clusters (Fig. 1c). Compared to flat surfaces, textured surfaces led to more clusters (for example, 7 versus 0 for 20 2PACz molecules on 6 nm wide SnO_2) (Supplementary Note 1 and Supplementary Figs. 3 and 4). The cluster formation is closely linked to the inhomogeneity of SAMs on FTO. For instance, when 162 2PACz molecules are positioned on top of textured SnO_2 surfaces, despite their initial random distribution (Supplementary Video 1), the final equilibrated structures exhibit clear phase segregation (Fig. 1d), in which almost 85% of the SnO_2 surface remains uncovered by any 2PACz molecules (Supplementary Fig. 5). These observations were also verified in a larger system (22 nm width; 1,134 2PACz molecules) (Supplementary Fig. 6).

Reasoning that a thiol group (–SH) could interact with phosphonic acids and a carboxyl group (–COOH) attached to FTO, we introduced 3-mercaptopropionic acid (3-MPA) (Fig. 1a) in simulations to break apart 2PACz clusters, particularly on textured substrates. The approach is akin to the co-adsorbent strategy used in dye-sensitized solar cells to

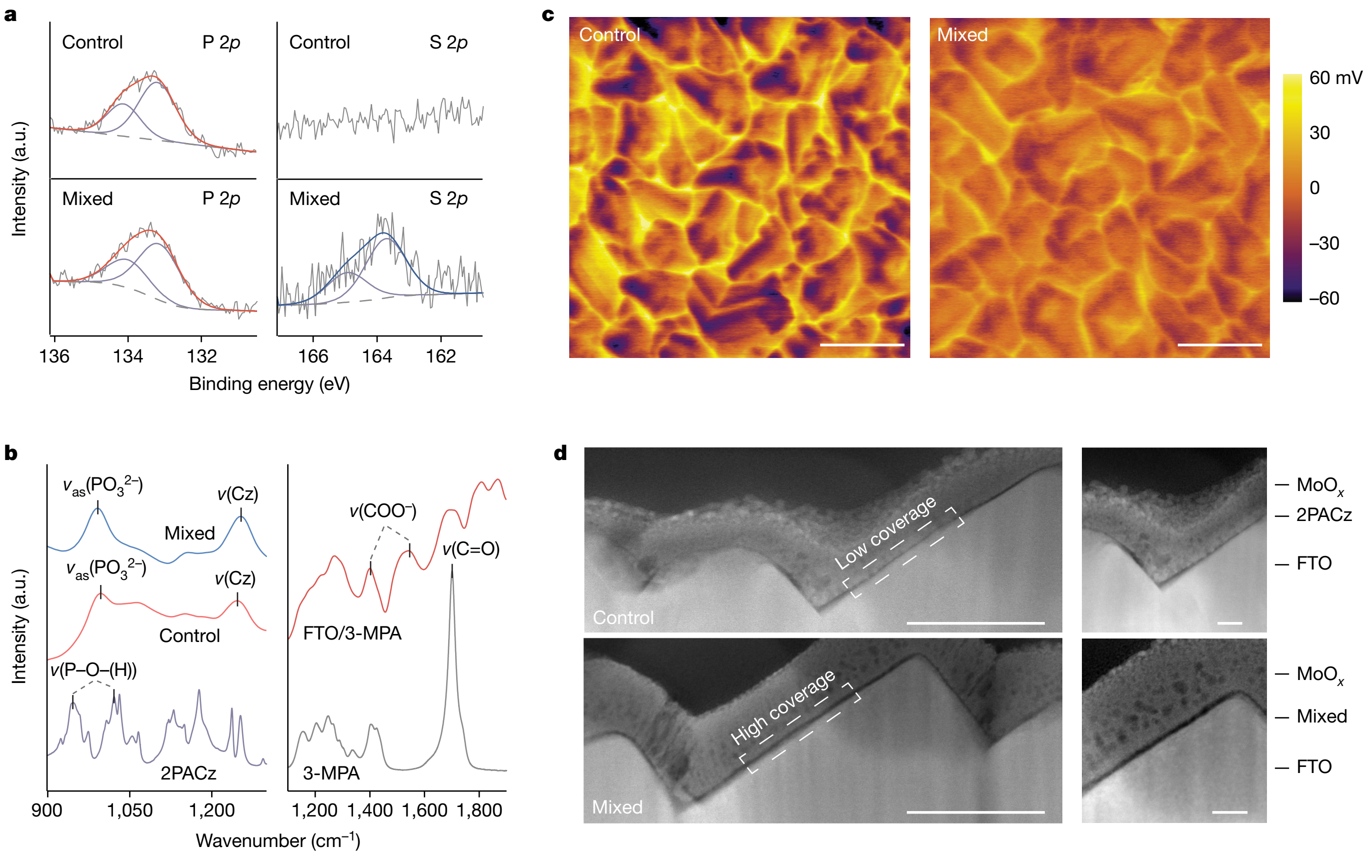

Fig. 2 | Homogeneity of self-assembled monolayers formed on FTO substrates. a, XPS P 2*p* (left) and S 2*p* (right) spectra for the control and mixed samples. The peaks were fitted using one S 2*p* or P 2*p* doublet with a 2:1 peak area ratio. **b**, 2PACz (left) and 3-MPA (right) transmission FTIR spectra, compared with ATR-FTIR spectra for the SAM-coated (left) and 3-MPA-coated (right) FTO substrates. The FTIR peaks identified at 947 and 1,021 cm^{-1} for 2PACz powders correspond to the symmetric and asymmetric (as) stretching of P–OH, respectively. Cz, carbazole. **c**, KPFM images of control (left) and mixed (right) SAM-coated FTO substrates. **d**, Cross-sectional HAADF-STEM images of control (top) and mixed (bottom) SAMs sandwiched between MoO_x and FTO. Scale bars, 500 nm (**c**); 100 nm (**d**, left); 20 nm (**d**, right).

reduce dye aggregation on titanium oxides[24]. Similar strategies have been used to improve the quality of hole-selective SAMs[25,26]. With 3-MPA at a molar ratio of 1:9 with 2PACz, we saw a relative 15% decrease in the number of clusters on textured surfaces (Fig. 1b), and a slower formation rate (20 ns^{-1} versus 28 ns^{-1}) (Supplementary Fig. 7). In particular, the higher order clusters (trimers and tetramers) decreased by 53% (Fig. 1c). As a result, the mixed system shows less phase segregation (Fig. 1e and Supplementary Video 2), leading to extended surface coverage (67% versus 15%) by 2PACz molecules (Supplementary Fig. 5). Ab initio molecular dynamics (AIMD) simulations (Supplementary Note 2 and Supplementary Fig. 8) indicate that the presence of 3-MPA hinders the free movement of individual 2PACz molecules by forming a supramolecular structure (Fig. 1f). This reduces agglomeration with already formed dimers, suppressing the formation of higher-order clusters (Supplementary Video 3). Density functional theory (DFT) calculations revealed surface binding energies of −3.2 eV and −2.6 eV for 2PACz and 3-MPA, respectively, which are much stronger than the interaction energies of 2PACz clusters (−0.3 to −0.1 eV per molecule). This indicates robust anchoring of 2PACz and 3-MPA molecules on contact with substrate surfaces.

Characterization of organic contacts

We deposited both 2PACz and a mixture of 2PACz:3-MPA on FTO substrates using solution processing, followed by annealing the film at 100 °C and solvent washing (Methods). Micelles were detected in the processing solutions through dynamic light scattering (DLS) (Supplementary Fig. 9). However, owing to their limited proportion in the volume distribution compared to 2PACz monomers, film formation primarily resulted from molecular diffusion rather than direct micellar adsorption[27].

The film composition and its interaction with FTO substrates were investigated using X-ray photoelectron spectroscopy (XPS) and attenuated total reflectance Fourier-transform infrared spectroscopy (ATR-FTIR), respectively. For the film consisting solely of 2PACz molecules (the control), the presence of 2PACz was evidenced by the appearance of the phosphine (P) doublet peak (Fig. 2a) and the characteristic C–N peak and C–C/C–H peak (Supplementary Fig. 10) in the XPS spectra[21]. In the case of mixed film, 3-MPA was indicated by the sulfur (S) 2*p* doublet peak at around 163.8 eV (that is, the thiol group) (Fig. 2a)[28]. From ATR-FTIR, bidentate or tridentate binding was identified as the mode of 2PACz–FTO interactions, given the appearance of the symmetric PO_3^{2-} stretching (at 996 cm^{-1} for the control sample)[29,30] and the disappearance of P–OH vibrations[14]. These also indicated that solvent washing removed unbound molecules[14]. With 3-MPA addition, a redshift of approximately 5 cm^{-1} was observed for the PO_3^{2-} vibrational mode, indicating the enhanced surface binding of 2PACz—as confirmed by AIMD simulations (Supplementary Note 3 and Supplementary Figs. 11 and 12). ATR-FTIR of the FTO/3-MPA sample (Fig. 2b) further showed carboxylate-related peaks and a reduction in the intensity of the C=O peak, which correspond to the bidentate chelation of 3-MPA with FTO surfaces[31]. Combining the results from XPS and ATR-FTIR, we reasoned that 3-MPA functioned as a co-adsorbent, modulating the interaction of 2PACz with FTO substrates.

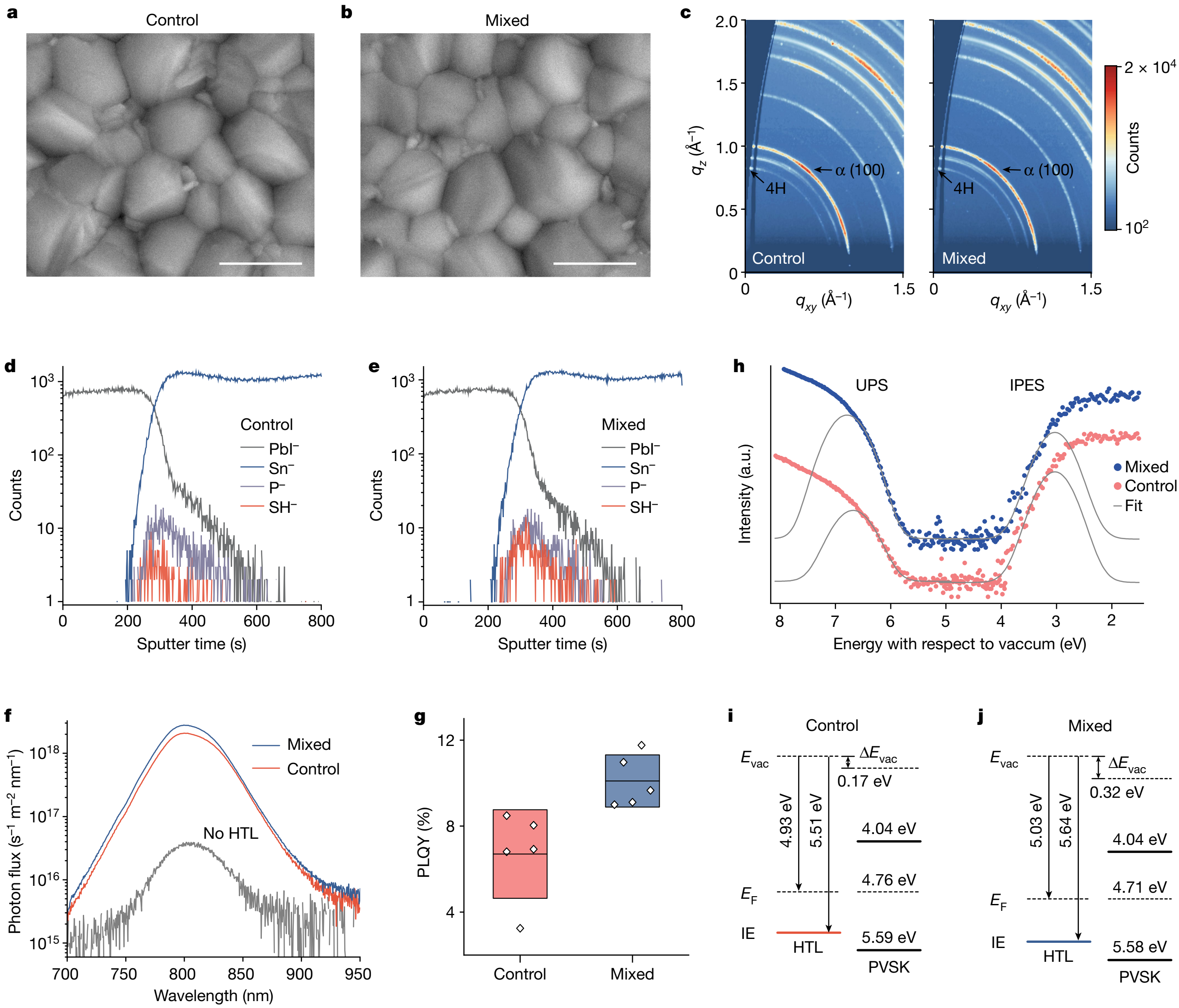

Fig. 3 | Materials properties of perovskite films on different FTO/SAM substrates. **a**,**b**, Top-view SEM images of perovskite films deposited on control (**a**) and mixed (**b**) SAMs. **c**, GIWAXS images for perovskite films on control (left) and mixed (right) SAMs. The colour bar shows the diffraction intensity collected from the GIWAXS detector. q_{xy} and q_z represent in-plane and near out-of-plane scattering vectors, respectively. **d**,**e**, TOF-SIMS profiles of perovskites on FTO/control SAM (**d**) and FTO/mixed SAM (**e**) substrates measured in negative polarity. Traces of SH⁻ were identified at the perovskite–control SAM interface, possibly owing to contamination of precursor materials. **f**, Absolute intensity photoluminescence spectra of perovskite films on bare FTO (no SAM) and control and mixed SAMs. **g**, PLQYs of perovskite films on control and mixed SAMs (five samples for each condition). Statistical distribution represented in box plots (line within the box: mean; box limit: standard deviation). **h**, UPS and IPES spectra of perovskite films on control and mixed SAMs. Gaussian fit (grey line) was used to determine the conduction and valence bands. **i**,**j**, Schematic energy-level diagrams of the perovskite/control SAM (**i**) and perovskite/mixed SAM bilayer (**j**), respectively. Scale bars, 1 µm. IE, ionization energy; HTL, hole-transport layer; PVSK, perovskite.

The resultant SAM uniformity was characterized by Kelvin probe force microscopy (KPFM). Topography images revealed that, despite varying SAM modifications, the surface morphology of FTO substrates was retained owing to the ultrathin nature of SAMs (Supplementary Fig. 13). From the respective contact potential difference (CPD) maps (Fig. 2c), we recorded surface potential variations on 2PACz-modified surfaces within a range of 120 mV. A Gaussian fit of the single peak in the CPD distribution yielded the full-width half-maximum (FWHM) of 37 mV (Supplementary Fig. 14). Introducing 3-MPA enhanced the homogeneity of electronic properties on SAM-modified surfaces, as reflected by a narrower CPD distribution with the FWHM of 22 mV.

To directly visualize the distribution of SAMs on FTO substrates, we performed high-angle annular dark-field (HAADF) scanning transmission electron microscopy (STEM) measurements. Figure 2d shows cross-sectional HAADF-STEM images of both control and mixed samples. Because of the contrast in the atomic number[2], SAM was discernible as a dark layer sandwiched between the brighter FTO and protective molybdenum oxide (MoO_x) layers. For the control SAM, we observed thickness variations across the same FTO facet, with certain regions showing thicknesses less than 1 nm. This suggests a low-density coverage and inhomogeneous distribution of 2PACz molecules[32]. The mixed sample exhibited significant improvements in terms of SAM uniformity and coverage; a consistent thickness was recorded for the mixed SAM on the same facet, albeit with variations between 1 and 2 nm across different facets. Correspondingly, mixed SAM-modified FTO substrates exhibited higher hydrophobicity than control SAM-modified counterparts (Supplementary Fig. 15). Cyclic voltammetry measurements further determined the areal density of 2PACz in the mixed SAM to be

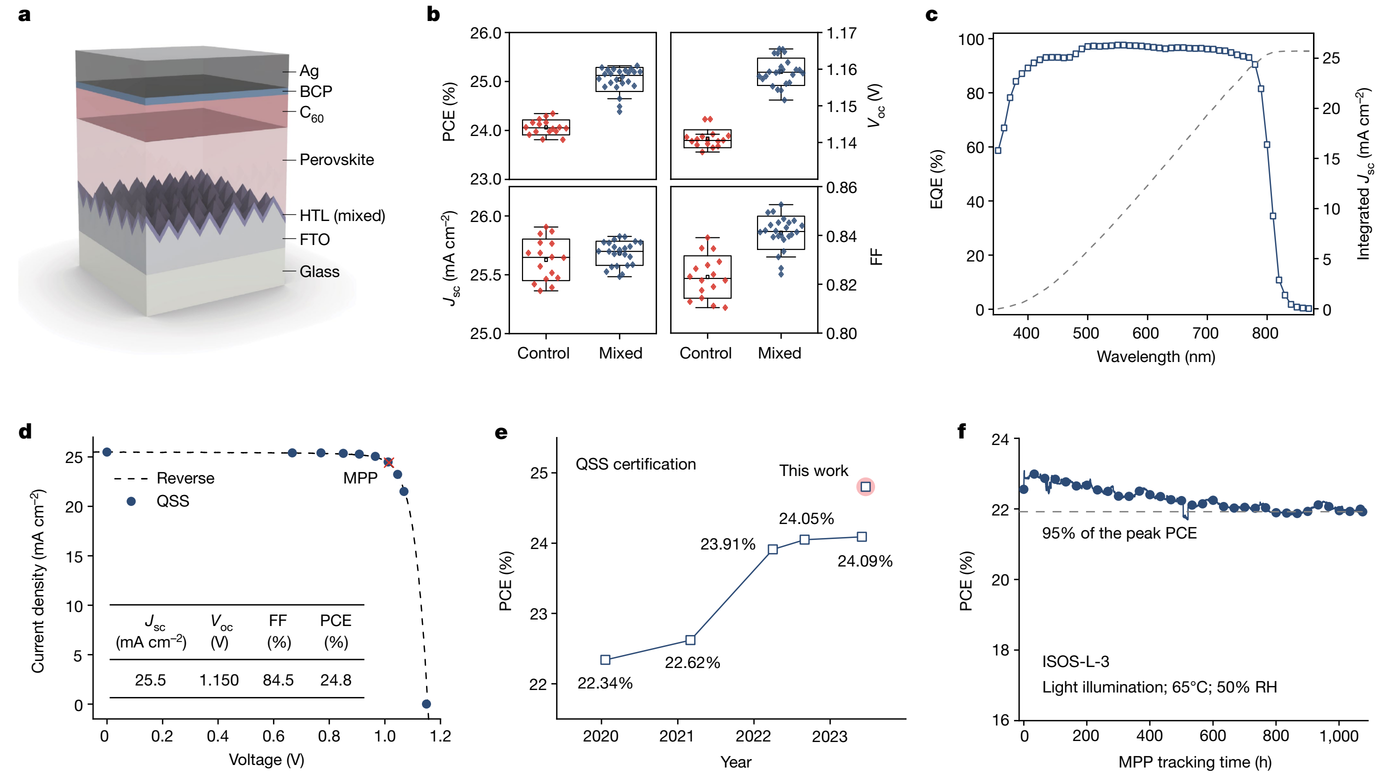

Fig. 4 | Photovoltaic performance of perovskite solar cells. a, Schematic illustration of the device architecture with textured FTO substrate. **b**, Solar cell parameters for the control (16 devices) and mixed (32 devices) SAM devices. Statistical distribution represented in box-and-whisker plots (line within box: median; box limit: standard deviation; whiskers: 1.5 × outliers). **c**, EQE and integrated J_{sc} (grey dashed line) curves of the mixed SAM device. **d**, QSS J–V curve of one representative mixed SAM device certified at Newport. Inset: PV parameters of the device. **e**, Certified performance of inverted PSCs measured under the QSS condition. **f**, MPP tracking of the encapsulated mixed SAM device at heat sink temperature of 65 °C with a relative humidity (RH) of approximately 50% under 1 sun illumination. The device configuration is FTO/SAM/perovskite/345FAn/C_{60}/ALD-SnO_2/Ag, with a PCE of 24.6% at room temperature. ALD, atomic-layer deposition; FF, fill factor.

3.9×10^{13} molecules per cm^2, a 70% increase from the 2.3×10^{13} molecules per cm^2 in the control SAM (Supplementary Fig. 16)[16,27].

Characterization of perovskite films

To investigate the impact of SAMs on the structural and optoelectronic properties of perovskites, we fabricated triple-cation $Cs_{0.05}MA_{0.10}FA_{0.85}PbI_3$ perovskite thin films on different FTO/SAM substrates. From scanning electron microscopy (SEM), we observed no notable change in the perovskite surface morphology when 3-MPA was incorporated into the SAM (Fig. 3a,b). Grazing incidence wide-angle X-ray scattering (GIWAXS) measurements showed nearly identical crystal structures for perovskites deposited on both the control and mixed SAMs (Fig. 3c). Here, the main constituent was photoactive α-phase perovskites, with traces of the 4H hexagonal phase detected at $q = 0.82$ Å^{-1} and 0.92 Å^{-1} as well as PbI_2 at $q = 0.90$ Å^{-1}, respectively (Supplementary Fig. 14)[33]. Time-of-flight secondary ion mass spectrometry (TOF-SIMS) further confirmed that both 3-MPA and 2PACz molecules, owing to their chemisorption on FTO surfaces, remained localized at the perovskite–FTO interfaces (Fig. 3d,e). These findings suggest that 3-MPA is not likely to change the perovskite formation process or contribute to the grain boundary or top surface passivation.

We conducted photoluminescence characterization by exciting perovskite thin films at a 1 sun equivalent photon flux. We saw an average photoluminescence quantum yield (PLQY) of 0.13% ± 0.02% for perovskites directly deposited onto FTO substrates, corresponding to a quasi-Fermi level splitting of 1.105 eV (Supplementary Table 1)[34]. 2PACz can passivate defects on metal oxide surfaces and effectively block electron injection[35], resulting in an approximately 50-fold increase in the photoluminescence intensity (Fig. 3f). We noted an average PLQY of 7% ± 2% for the perovskite/2PACz/FTO stacks (Fig. 3g), corresponding to a quasi-Fermi level splitting of 1.207 eV. The incorporation of 3-MPA further improved the average PLQY to 10% ± 1%. This we ascribed to better SAM coverage rather than 3-MPA passivation, as the thiol group of 3-MPA was embedded within the SAM (Fig. 1f).

We used ultraviolet and inverse photoelectron spectroscopy (UPS and IPES) to determine the work function and transport energy levels of perovskites and SAMs. The secondary electron cutoff in the UPS spectra indicated an increase in the work function from 4.56 to 4.93 eV on depositing the control SAM on the FTO substrate (Supplementary Fig. 18). 3-MPA led to an additional work function shift of +100 meV. The ionization energy was 5.51 and 5.64 eV for the control and mixed SAMs, respectively. By contrast, perovskites exhibited similar conduction band minima and valance band maxima on different substrates (Fig. 3h). The resultant energy-level diagrams of perovskites and SAMs are depicted in Fig. 3i,j. A vacuum level shift of 0.17 eV was recorded between the perovskite and the 2PACz bilayer, which further increased to 0.32 eV for the perovskite/mixed SAM bilayer[36]. A higher vacuum level shift indicates an enlarged built-in field[37], which in PSCs can lead to enhanced carrier extraction[38]. Meanwhile, the ionization energy offset between the bilayer remained consistent for the control and mixed SAMs (referenced to the Fermi level (E_F)).

Solar cell efficiency and stability

We fabricated PSCs with the following inverted structure: FTO/SAM/perovskites/3,4,5-trifluoroanilinium (345FAn)/fullerene (C_{60})/bathocuproine (BCP)/Ag (Fig. 4a and Supplementary Fig. 19). 345FAn was selected for interface engineering because of its thermal stability[1]. 2PACz (control) and 2PACz:3-MPA (mixed) were used as the SAMs for hole-selective contact. The corresponding solar cell parameters are summarized in Fig. 4b. We found that compared to control devices, the mixed SAM led to improved performance (average PCE 25.0% compared with 24.1%). This came from improved open-circuit voltage (V_{oc}, from 1.141 V to 1.159 V) and fill factor (from 82.3% to 84.1%) (Supplementary Note 4). We noted that device reproducibility was comparable for both control and mixed SAMs, probably owing to SAM inhomogeneity occurring at the nanoscale[18,27]. The champion device with the mixed SAM exhibited a PCE of 25.3% from the reverse *J*–*V* scan, which agrees with the PCE obtained from the steady-state power output (Supplementary Fig. 20). The integrated short-circuit current density (J_{sc}) of 25.8 mA cm^{-2} derived from the external quantum efficiency (EQE) measurement matches well with that from the *J*–*V* sweep (Fig. 4c).

One mixed SAM-based FTO device was characterized at Newport; and produced a QSS-certified PCE of 24.8%, with a V_{oc} of 1.150 V, a J_{sc} of 25.5 mA cm^{-2} and a fill factor of 84.5% (Fig. 4d and Supplementary Fig. 21). Although there have been reports of inverted PSCs with efficiencies surpassing 25% (refs. 2,3), the certified PCEs measured under stabilized conditions (including maximum power point (MPP) and QSS tracking) have yet to reach the same level (Supplementary Table 2). The QSS efficiency reported herein represents a new record for inverted PSCs, improving on the previous record QSS efficiency of 24.09% in the literature (Fig. 4e).

For comparison, inverted PSCs were fabricated on smooth ITO substrates. We obtained an average PCE of 23.4% for control ITO devices, which was lower than that of control FTO devices, despite their higher average V_{oc} and fill factor (Supplementary Fig. 22). This emphasizes the need to augment J_{sc} for inverted PSCs. The limited J_{sc} is ascribed to the smoothness of ITO substrates (Supplementary Fig. 23), leading to reduced light scattering and thus insufficient light absorption in the full device (Supplementary Fig. 24). Interestingly, the performance of mixed SAM-based ITO devices was only slightly improved, possibly owing to the weak interactions of 3-MPA with ITO surfaces (Supplementary Note 5 and Supplementary Fig. 25).

We evaluated the operating stability of PSCs using International Summit on Organic Photovoltaic Stability (ISOS)-L-3 protocols, in which the encapsulated device was subjected to continuous 1 sun equivalent, white light-emitting diode illumination (Supplementary Fig. 26) at 50% relative humidity and heat sink temperature of 65 °C. To impede ion and moisture diffusion, we replaced BCP with atomic layer-deposited SnO_2 as a buffer layer[9]. The resultant device delivered a PCE of 24.6% at room temperature (Supplementary Fig. 27). During the ISOS-L-3 testing, the device initially demonstrated a PCE of 22.6%, which increased to 23.1% after 1.6 h of MPP tracking (Fig. 4f). The lower PCE at 65 °C compared to room temperature is ascribed to the negative temperature coefficient of PSCs (−0.15% per °C)[39]. The PCE stabilized at 21.9% (95% of the peak PCE) until the end of the test (1,075 h), with the main degradation in the photocurrent (Supplementary Fig. 28). The initial performance and operating stability reported herein are compared with other PSCs subjected to ISOS-L-3 tests (Supplementary Table 3).

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-023-06745-7.

1. Park, S. M. et al. Engineering ligand reactivity enables high-temperature operation of stable perovskite solar cells. *Science* **381**, 209–215 (2023).
2. Zhang, S. et al. Minimizing buried interfacial defects for efficient inverted perovskite solar cells. *Science* **380**, 404–409 (2023).
3. Jiang, Q. et al. Surface reaction for efficient and stable inverted perovskite solar cells. *Nature* **611**, 278–283 (2022).
4. Kim, M. et al. Conformal quantum dot–SnO_2 layers as electron transporters for efficient perovskite solar cells. *Science* **375**, 302–306 (2022).
5. Park, J. et al. Controlled growth of perovskite layers with volatile alkylammonium chlorides. *Nature* **616**, 724–730 (2023).
6. Best Research-Cell Efficiency Chart. *NREL* www.nrel.gov/pv/cell-efficiency.html (2023).
7. You, S. et al. Radical polymeric p-doping and grain modulation for stable, efficient perovskite solar modules. *Science* **379**, 288–294 (2023).
8. Zheng, X. et al. Managing grains and interfaces via ligand anchoring enables 22.3%-efficiency inverted perovskite solar cells. *Nat. Energy* **5**, 131–140 (2020).
9. Chen, H. et al. Quantum-size-tuned heterostructures enable efficient and stable inverted perovskite solar cells. *Nat. Photonics* **16**, 352–358 (2022).
10. Li, Z. et al. Organometallic-functionalized interfaces for highly efficient inverted perovskite solar cells. *Science* **376**, 416–420 (2022).
11. Peng, W. et al. Reducing nonradiative recombination in perovskite solar cells with a porous insulator contact. *Science* **379**, 683–690 (2023).
12. Camaioni, N. et al. Polymer solar cells with active layer thickness compatible with scalable fabrication processes: a meta-analysis. *Adv. Mater.* **35**, 2210146 (2023).
13. Sahli, F. et al. Fully textured monolithic perovskite/silicon tandem solar cells with 25.2% power conversion efficiency. *Nat. Mater.* **17**, 820–826 (2018).
14. Al-Ashouri, A. et al. Conformal monolayer contacts with lossless interfaces for perovskite single junction and monolithic tandem solar cells. *Energy Environ. Sci.* **12**, 3356–3369 (2019).
15. Paniagua, S. A. et al. Phosphonic acids for interfacial engineering of transparent conductive oxides. *Chem. Rev.* **116**, 7117–7158 (2016).
16. Truong, M. A. et al. Tripodal triazatruxene derivative as a face-on oriented hole-collecting monolayer for efficient and stable inverted perovskite solar cells. *J. Am. Chem. Soc.* **145**, 7528–7539 (2023).
17. Isikgor, F. H. et al. Molecular engineering of contact interfaces for high-performance perovskite solar cells. *Nat. Rev. Mater.* **8**, 89–108 (2022).
18. Mao, L. et al. Fully textured, production-line compatible monolithic perovskite/silicon tandem solar cells approaching 29% efficiency. *Adv. Mater.* **34**, e2206193 (2022).
19. He, R. et al. Improving interface quality for 1-cm^2 all-perovskite tandem solar cells. *Nature* **618**, 80–86 (2023).
20. Phung, N. et al. Enhanced self-assembled monolayer surface coverage by ALD NiO in p-i-n perovskite solar cells. *ACS Appl. Mater. Interfaces* **14**, 2166–2176 (2022).
21. Farag, A. et al. Evaporated self-assembled monolayer hole transport layers: lossless interfaces in p-i-n perovskite solar cells. *Adv. Energy Mater.* **13**, 2203982 (2023).
22. Boyd, C. C. et al. Overcoming redox reactions at perovskite-nickel oxide interfaces to boost voltages in perovskite solar cells. *Joule* **4**, 1759–1775 (2020).
23. Fukano, T. Low-temperature growth of highly crystallized transparent conductive fluorine-doped tin oxide films by intermittent spray pyrolysis deposition. *Sol. Energy Mater. Sol. Cells* **82**, 567–575 (2004).
24. Neale, N. R., Kopidakis, N., van de Lagemaat, J., Grätzel, M. & Frank, A. J. Effect of a coadsorbent on the performance of dye-sensitized TiO_2 solar cells: shielding versus band-edge movement. *J. Phys. Chem. B* **109**, 23183–23189 (2005).
25. Deng, X. et al. Co-assembled monolayers as hole-selective contact for high-performance inverted perovskite solar cells with optimized recombination loss and long-term stability. *Angew. Chem. Int. Ed. Engl.* **61**, e202203088 (2022).
26. Al-Ashouri, A. et al. Wettability improvement of a carbazole-based hole-selective monolayer for reproducible perovskite solar cells. *ACS Energy Lett.* **8**, 898–900 (2023).
27. Liu, M. et al. Compact hole-selective self-assembled monolayers enabled by disassembling micelles in solution for efficient perovskite solar cells. *Adv. Mater.* **35**, 2304415 (2023).
28. Love, J. C., Estroff, L. A., Kriebel, J. K., Nuzzo, R. G. & Whitesides, G. M. Self-assembled monolayers of thiolates on metals as a form of nanotechnology. *Chem. Rev.* **105**, 1103–1170 (2005).
29. Paniagua, S. A. et al. Phosphonic acid modification of indium–tin oxide electrodes: combined XPS/UPS/contact angle studies. *J. Phys. Chem. C* **112**, 7809–7817 (2008).
30. Hotchkiss, P. J., Malicki, M., Giordano, A. J., Armstrong, N. R. & Marder, S. R. Characterization of phosphonic acid binding to zinc oxide. *J. Mater. Chem.* **21**, 3107 (2011).
31. Nazeeruddin, M. K., Humphry-Baker, R., Liska, P. & Grätzel, M. Investigation of sensitizer adsorption and the influence of protons on current and voltage of a dye-sensitized nanocrystalline TiO_2 solar cell. *J. Phys. Chem. B* **107**, 8981–8987 (2003).
32. Bunker, B. C. et al. The impact of solution agglomeration on the deposition of self-assembled monolayers. *Langmuir* **16**, 7742–7751 (2000).
33. Dang, H. X. et al. Multi-cation synergy suppresses phase segregation in mixed-halide perovskites. *Joule* **3**, 1746–1764 (2019).
34. Caprioglio, P. et al. On the relation between the open-circuit voltage and quasi-Fermi level splitting in efficient perovskite solar cells. *Adv. Energy Mater.* **9**, 1901631 (2019).
35. Al-Ashouri, A. et al. Monolithic perovskite/silicon tandem solar cell with >29% efficiency by enhanced hole extraction. *Science* **370**, 1300–1309 (2020).
36. Li, X. et al. Mapping the energy level alignment at donor/acceptor interfaces in non-fullerene organic solar cells. *Nat. Commun.* **13**, 2046 (2022).
37. Lami, V. et al. Visualizing the vertical energetic landscape in organic photovoltaics. *Joule* **3**, 2513–2534 (2019).
38. Sandberg, O. J. et al. On the question of the need for a built-in potential in perovskite solar cells. *Adv. Mater. Interfaces* **7**, 2000041 (2020).
39. Moot, T. et al. Temperature coefficients of perovskite photovoltaics for energy yield calculations. *ACS Energy Lett.* **6**, 2038–2047 (2021).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Methods

Molecular dynamics simulations

Static DFT calculation and AIMD were performed with the CP2K[40] package utilizing a mixed Gaussian/plane wave (GPW) basis. The PBE exchange-correlation functional[41], DFT-D3 dispersion corrections[42] and dipole corrections[43] necessary for periodic boundary conditions along the perpendicular direction to the surface were included. Valence electron wave functions were expanded in a double-ζ Gaussian basis set with polarization functions (DZVP)[44]. The energy cutoff for the electron density expansion in the GPW method was 400 Ry. Born–Oppenheimer AIMD simulations were run with an integration time step of 0.5 fs and the system was kept at 300 K using the thermostat of Bussi et al.[45] in a canonical isothermal–isochoric (NVT) ensemble, in which the total number of atoms N, the volume V and the temperature T of the system were held constant. All AIMD simulations were performed at Γ point. Being the most stable surface, the SnO_2(110) was chosen as the substrate surface. The systems were equilibrated over 5 ps in the NVT ensemble and the remaining 40 ps were used for the production run. To investigate the interactions between the 2PACz and 3-MPA molecules, as well as molecules and the SnO_2 substrate, five scenarios were simulated: (1) a single molecule (2PCAz or 3-MPA) on SnO_2(110); (2) one 2PACz and one 3-MPA concurrently on SnO_2(110); (3) two 2PACz and one 3-MPA on SnO_2(110); (4) four 2PACz and one 3-MPA on SnO_2(110); and (5) six 2PACz and one 3-MPA on SnO_2(110). In all the scenarios, an in-plane SnO_2(110) simulation cell of 22.690 Å × 27.756 Å with 45 Å of vacuum between the slab repetitions was used. AIMD was also used to estimate the interaction energy of 2PACz and 3-MPA with the SnO_2(110) surface. The final AIMD configurations were relaxed with DFT to extract the interaction energies of the molecules. Binding energies were calculated as $\Delta E_b = E_{tot} - E_S - E_{mol}$, where E_{tot} is the energy of the final relaxed configuration (molecule on SnO_2(110)); E_S is the energy of the final configuration of the slab; and E_{mol} is the energy of the final configuration of the isolated molecule. The interaction energies of 2PACz clusters were computed after DFT relaxations of isolated clusters. Dimers, trimers and tetramers were considered. Interaction energies per molecule were calculated as $\Delta E_i = (E_{tot} - nE_{mol})/n$, where E_{tot} is the energy of the final relaxed cluster; n is the number of molecules forming the cluster; and E_{mol} is the energy of a single relaxed molecule assumed isolated. Vibrational power spectra were computed as the Fourier transform of the autocorrelation function of the atomic velocities.

Classical molecular dynamics simulations with full atomic resolution were realized with the help of the LAMMPS molecular dynamics simulator[46]. The equations of motion were integrated using the velocity Verlet method[47], with a 1 fs time step, in which bond stretching was not constrained for any bonds. To maintain isothermal conditions, the deterministic Nosé–Hoover thermostat[48,49], with a time constant equal to 0.1 ps, was used. This ensured that thermostating was applied to both translational and rotational degrees of freedom, crucial for the asymmetric 2PACz and 3-MPA molecules. All classical molecular dynamics simulations were realized in the canonical NVT ensemble at T = 300 K comprising a V-shaped SnO_2(110) model system with dimensions 60 Å × 105 Å × 40 Å. Periodic boundary conditions were used in x and y directions, whereas reflective boundary conditions were used in the z direction. The two considered systems, control and mixed, consisted of a total of 18,498 and 18,714 atoms, respectively, whereas the experimental molar ratio (2PACz:3-MPA = 9:1) was retained in the simulated mixed system. The systems were equilibrated for 1 ns followed by a 10 ns production simulation, during which sampling took place every 500 fs for the calculation of ensemble averages. A snapshot of the system was taken every 1,000 fs for visualization purposes. The calculation of the number distributions of formed 2PACz clusters was performed through an in-house code using a threshold distance of 8 Å among the centre of masses of adjacent 2PACz molecules, in accordance with the nearest neighbour shell of the corresponding radial distribution function as clustering criterion.

To accurately capture the interactions of SnO_2, the thoroughly validated classical force field developed by Bandura et al.[50] was used. The all-atom OPLS force field was utilized for 3-MPA, wherein the corresponding interaction parameters were obtained with the help of the LigParGen server[51]. For 2PACz, containing the phosphonic functional group, the specially designed force field developed by Meltzer et al.[52] based on the generalized Amber force field[53] was used. The van der Waals and electrostatic non-bonded interactions were calculated using a real-space cutoff radius of 10 Å and 8 Å, respectively, whereas the particle–particle–particle–mesh scheme[54] with an accuracy of 0.0001 (kcal mol^{-1}) Å^{-1} was used for the consideration of long-range electrostatic interactions. Because the SnO_2 force field uses a Buckingham potential for the van der Waals interactions, its combination with the Lennard-Jones potential used for the compounds was realized by fitting the Buckingham parameters of SnO_2 to a suitable set of Lennard-Jones parameters[55] and geometric mixing rules were applied for all dissimilar non-bonded interactions.

Materials

All materials were used as received without further purification. Organic halide salts, including methylammonium iodide (MAI), formamidinium iodide (FAI) and methylammonium chloride (MACl), were purchased from Great Cell Solar. Caesium iodide (CsI), 3-mercaptopropionic acid (3-MPA), 3,4,5-trifluoroaniline (98%) and guanidinium thiocyanate (GuaSCN) were purchased from Sigma-Aldrich. 3,4,5-Trifluoroaniline was converted to its ammonium form following the published procedure. 2PACz, lead iodide (PbI_2, 99.99%) and BCP were purchased from TCI. Anhydrous solvents, including N,N-dimethylformamide (DMF, 99.8%), dimethyl sulfoxide (DMSO, 99.9%), 2-propanol (IPA, 99.5%), chloroform (CF, 99.8%) and anisole (99.7%) were purchased from Sigma-Aldrich. The C_{60} was purchased from Xi'an Polymer Light Technology. ITO and FTO (TEC 10) substrates were purchased from Thin Film Devices and Ossila, respectively.

Perovskite film fabrication

First, 1.5 M $Cs_{0.05}MA_{0.1}FA_{0.85}PbI_3$ perovskite precursor was fully dissolved in mixed solvents of DMF and DMSO (4:1, v/v) with the molar ratio for FAI/MAI/CsI of 0.85:0.1:0.05. Then, 10 mg ml^{-1} MACl and 8 mg ml^{-1} GuaSCN were added to the solution to improve the film morphology. The precursor solution was filtered through a 0.22 µm polytetrafluoroethylene membrane before use. A portion of 60 µl of perovskite solution was deposited on the substrate and spun cast at 1,000 rpm for 10 s, followed by 6,000 rpm for 30 s. A 150 µl portion of anisole was dropped onto the substrate during the last 5 s of spinning, resulting in the formation of dark brown films that were then annealed on a hot plate at 100 °C for ITO and 120 °C for FTO, for 20 min.

Device fabrication

FTO glasses were sequentially sonicated in aqueous detergent, deionized water, acetone and IPA each for 10 min. After drying with nitrogen, the substrates were exposed to UV–ozone treatment for 15 min to remove organic contaminants and then transferred into a nitrogen-filled glovebox maintaining less than 0.1 ppm of O_2 and H_2O. Then, 100 µl of 2PACz or a mixture of 2PACz:3-MPA (in a molar ratio of either 9:1 in characterization and device fabrication, or 8:2 in XPS measurements) in anhydrous ethanol (1 mM) solution was uniformly spread on the substrates and allowed to rest for 15 s, followed by spinning of the films at 3,000 rpm for 30 s. The films were then annealed at 100 °C for 10 min and washed by dropping 500 µl ethanol during the film spinning at 3,000 rpm. Perovskite solution was deposited on the SAM-modified substrate as detailed above. A 200 µl portion of 345FAn solution (1 mg ml^{-1}) in CF with an additional 3% of IPA was then drop cast within 2–3 s on the perovskite film spinning at 4,000 rpm (that is, dynamic spinning) and annealed at 100 °C for 5 min. Both control and mixed SAM films were then transferred to the thermal evaporator (Angstrom Engineering). The C_{60} (30 nm) and BCP (7 nm) were deposited sequentially with a rate of 0.3 Å s^{-1} and 0.5 Å s^{-1}, respectively, at a pressure of approximately 2×10^{-6} mbar. Finally, Ag contact (140 nm) was deposited on top of BCP through a shadow mask with the desired aperture area. For the stability testing, ALD-SnO_2 was used as the barrier layer. The deposition of ALD-SnO_2 was carried out in the PICOSUN R-200 Advanced ALD system. H_2O and TDMASn were used as oxygen and tin precursors. Precursor and substrate temperature were set to 75 °C and 85 °C, respectively. The carrier gas used was 90 sccm N_2. Pulse and purge times for H_2O were 1 s and 5 s, and 1.6 s and 5 s for TDMASn. The total deposition cycle was 120, corresponding to 20 nm of SnO_2.

Solar cell characterization

The current–voltage (*J*–*V*) characteristics of solar cells were measured using a Keithley 2400 source meter under the illumination of a solar simulator (Newport, Class AAA) at a light intensity of 100 mW cm^{-2}, as checked using a calibrated reference solar cell (Newport). The stabilized PCE was measured by setting the bias voltage to the V_{MPP} and then tracing the current density. The V_{MPP} was determined from the reverse *J*–*V* curve. The active area was determined by the aperture shade mask (0.049 cm^2) placed in front of the solar cell to avoid overestimation of the photocurrent density. EQE spectra were recorded with a commercial system (Arkeo-Ariadne, Cicci Research) based on a 300 W xenon light source and a holographic grating monochromator (Cornerstone, Newport).

Stability tests of solar cells

Devices were placed in a homemade stability tracking station. The illumination source was a white-light LED with intensity calibrated to match the 1 sun condition. For the ISOS-L-3 ageing protocol (65 or 85 °C; 50% relative humidity; MPP)[56], the device chamber was left open in a room with 50 ± 10% humidity and the solar cell was mounted on a metal plate kept at 65 °C by a heating element. A thermal couple attached to the metal plate was used to monitor and provide feedback control to the heating element to ensure temperature consistency. MPP was tracked using a home-built MATLAB-based MPP tracking system using a ‘perturb and observe’ method. The MPP was updated every 1,000 min. Encapsulation was done by capping the device with a glass slide, using UV adhesive (Lumtec LT-U001) as a sealant.

XPS measurements

XPS measurements were performed with a Thermo Scientific K-Alpha system with 180° double-focusing, hemispherical analyser. The system is equipped with a 128-channel detector and monochromatic small-spot XPS. An AlKα source (1,486.6 eV) was used for excitation and a pass energy of 50 eV was used for XPS acquisition. Samples were mounted on a metal specimen holder. All data were analysed with CasaXPS and Thermo Avantage software.

KPFM measurements

KPFM measurements were performed using an Asylum Research Cypher S atomic force microscope (Oxford Instruments) with an ASYELEC.01-R2 Ti-Ir coated cantilever (Asylum Research). Scans were performed over 2 µm at 512 pixels and 0.5 Hz using a two-pass method in which the first pass is a tapping mode topography scan and the second is in KPFM mode with a tip potential of 3 V and a surface clearance of 5 nm. Cantilever calibration was performed using the Thermal method from the Asylum Research GetReal database.

HAADF-TEM measurements

HAADF-TEM images were acquired at an aberration-corrected FEI Titan Cubed Themis G2 operated at 300 kV equipped with an XFEG gun and Bruker Super-X EDS detectors. The SAM samples had a structure of glass/FTO/SAM/MoO_x (50 nm), in which the MoO_x layer was deposited through thermal evaporation with a low rate of 0.1 Å s^{-1} at a pressure below 2×10^{-4} Pa. The cross-sectional samples were prepared by using a focused ion-beam system (Helios G4 UX). Another protective layer of carbon was thermally evaporated before ion-beam cutting and etching.

Cyclic voltammetry measurements

Cyclic voltammetry measurements were conducted using a three-electrode configuration with a potentiostat (PGSTAT204, Autolab). The working electrodes were prepared using a spin-coating method on an FTO electrode. The exposed area of the working electrode to the electrolyte measured 8 mm × 21.4 mm. A platinum plate and an Ag/AgCl electrode (in a 3.0 M KCl solution) were used as the counter and reference electrodes, respectively. The measurements were performed in an Ar-saturated solution of 1,2-dichlorobenzene (*o*-DCB) with 0.1 M tetrabutylammonium hexafluorophosphate ($TBA^+PF_6^-$) serving as the supporting electrolyte. All potentials were referenced against the ferrocene redox couple, serving as an internal standard. The effective coverage of the self-assembled monolayers on the FTO surface was measured by the slope of a linear dependency of the oxidative peak intensity against the scan rate, as follows:

$$i_{p,o} = \frac{n^2F^2}{4RTN_A} A\Gamma^* \upsilon$$

where $i_{p,o}$ is the oxidative peak current, υ is the voltage scan rate, n is the number of electrons transferred, F is the Faraday constant (96,485 C mol^{-1}), R is the universal gas constant (8.314 J K^{-1} mol^{-1}), T is the temperature, N_A is the Avogadro constant, A is the electrode area and Γ^* is the areal density.

UPS and IPES measurements

UPS measurements were taken with an Excitech H Lyman-α photon source (10.2 eV) with a nitrogen-filled beam path coupled with a PHI 5600 UHV and analyser system. A sample bias of −5 V was applied and a pass energy of 5.85 eV was used for UPS acquisition. IPES measurements were performed in the Bremsstrahlung isochromatic mode with electron kinetic energies below 5 eV and an electron gun emission current of 2 µA was used to minimize sample damage. A Kimball Physics ELG-2 electron gun with a BaO cathode was used to generate the electron beam. Emitted photons were collected with a bandpass photon detector consisting of an optical bandpass filter (254 nm, Semrock) and a photomultiplier tube (R585, Hamamatsu Photonics). Samples were held at a −20 V bias during all IPES measurements and the UHV chamber was completely dark.

Other characterizations

Particle size distributions in solutions were determined by the DLS technique using a Malvern Zetasizer Nano ZS. Contact angles were measured using VCA-Optima XE. The image was taken with a CCD camera within 1 s after water droplet. GIWAXS measurements were performed at the CMS beamline, NSLS II. The monochromatic X-ray with an energy of 13.5 keV was shone on the samples at different grazing incident angles of 1°, 0.5° and 0.08° with an exposure time of 10 s. A Pilatus 800K detector was placed 259 mm away from the sample to capture the two-dimensional diffraction pattern. Absolute intensity photoluminescence spectra were measured using an integrating sphere, and Andor Kymera 193i spectrograph, and a 660 nm continuous-wave laser set at 1 sun equivalent photon flux (1.1 µm beam FWHM, 632 µW); photoluminescence was collected at normal incidence using a 0.1 numerical aperture, 110-µm-diameter optical fibre. TOF-SIMS was conducted on the IONTOF M6 instrument with a Bi^{3+} (30 keV) primary ion beam for analysis and a Cs cluster gun (2 keV) for sputtering. Data were acquired in positive mode with an analysis area of 49 × 49 µm^2 centred and a raster area of 200 × 200 µm^2. IR spectra were obtained in the attenuated total reflectance mode using a Fourier-transform IR spectrometer (Thermo Scientific iS50). Samples were prepared on the FTO substrate and scanned in the spectral range of 4,000 to 550 cm^{-1} with a minimum number of 500 scans and a resolution of 4 cm^{-1}. The triangular apodization function was used to improve the signal-to-noise ratio. High-resolution SEM images were obtained using the Hitachi S5200 microscope with an accelerating voltage of 1.5 kV. A low accelerating voltage and a low beam current were deployed to reduce surface damage of perovskite films under electron beam bombardment. The diffuse and specular light reflected from the substrate surface was measured using a Cary 5000 UV-Vis-NIR double-beam spectrophotometer in diffuse reflectance mode. Pure $BaSO_4$ was used for the baseline collection.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data are available in the main text or the supplementary materials. Further data are available from the corresponding author on reasonable request.

Code availability

The codes and postanalysis tools for molecular dynamics simulations are available in the following repository: https://doi.org/10.5281/zenodo.8393081.

40. Kühne, T. D. et al. CP2K: an electronic structure and molecular dynamics software package—Quickstep: efficient and accurate electronic structure calculations. *J. Chem. Phys.* **152**, 194103 (2020).
41. Van Setten, M. J. et al. The PseudoDojo: training and grading a 85 element optimized norm-conserving pseudopotential table. *Comput. Phys. Commun.* **226**, 39–54 (2018).
42. Grimme, S. Density functional theory with London dispersion corrections. *WIREs Comput. Mol. Sci.* **1**, 211–228 (2011).
43. Bengtsson, L. Dipole correction for surface supercell calculations. *Phys. Rev. B* **59**, 12301–12304 (1999).
44. VandeVondele, J. & Hutter, J. Gaussian basis sets for accurate calculations on molecular systems in gas and condensed phases. *J. Chem. Phys.* **127**, 114105 (2007).
45. Bussi, G., Donadio, D. & Parrinello, M. Canonical sampling through velocity rescaling. *J. Chem. Phys.* **126**, 014101 (2007).
46. Kryuchkov, N. P., Yurchenko, S. O., Fomin, Y. D., Tsiok, E. N. & Ryzhov, V. N. Complex crystalline structures in a two-dimensional core-softened system. *Soft Matter* **14**, 2152–2162 (2018).
47. Swope, W. C., Andersen, H. C., Berens, P. H. & Wilson, K. R. A computer simulation method for the calculation of equilibrium constants for the formation of physical clusters of molecules: application to small water clusters. *J. Chem. Phys.* **76**, 637–649 (1982).
48. Nosé, S. A unified formulation of the constant temperature molecular dynamics methods. *J. Chem. Phys.* **81**, 511–519 (1984).
49. Hoover, W. G. Canonical Dynamics: Equilibrium Phase-Space Distributions. *Phys. Rev. A* **31**, 1695–1697 (1985).
50. Bandura, A. V., Sofo, J. O. & Kubicki, J. D. derivation of force field parameters for SnO_2–H_2O surface systems from plane-wave density functional theory calculations. *J. Phys. Chem. B* **110**, 8386–8397 (2006).
51. Dodda, L. S., Cabeza de Vaca, I., Tirado-Rives, J. & Jorgensen, W. L. LigParGen web server: an automatic OPLS-AA parameter generator for organic ligands. *Nucleic Acids Res.* **45**, W331–W336 (2017).
52. Meltzer, C. et al. indentation and self-healing mechanisms of a self-assembled monolayer—a combined experimental and modeling study. *J. Am. Chem. Soc.* **136**, 10718–10727 (2014).
53. Wang, J., Wolf, R. M., Caldwell, J. W., Kollman, P. A. & Case, D. A. Development and testing of a general Amber force field. *J. Comput. Chem.* **25**, 1157–1174 (2004).
54. Hockney, R. W. & Eastwood, J. W. *Computer Simulation Using Particles* (CRC Press, 2021); https://doi.org/10.1201/9780367806934.
55. Lim, T.-C. Alignment of Buckingham parameters to generalized Lennard-Jones potential functions. *Z. Naturforsch. A* **64**, 200–204 (2009).
56. Khenkin, M. V. et al. Consensus statement for stability assessment and reporting for perovskite photovoltaics based on ISOS procedures. *Nat. Energy* **5**, 35–49 (2020).

Acknowledgements This research was made possible by the US Department of the Navy, Office of Naval Research Grant (N00014-20-1-2572). This work was supported in part by Ontario Research Fund-Research Excellence programme (ORF7-Ministry of Research and Innovation, Ontario Research Fund-Research Excellence Round 7). This work was also supported under award number OSR-CRG2020-4350.2. M.W. acknowledges funding from the European Union's Horizon 2020 Research and Innovation programme under the Marie Skłodowska-Curie grant agreement no. 101026353. T.H., H.R.A. and K.R.G. gratefully acknowledge funding from the National Science Foundation under award no. DMR-2102257. L.X. acknowledges support by National Natural Science Foundation of China (no. 61935016, 52173153) and the Electron Microscopy Laboratory of Peking University for the use of electron microscopes. M.G.K. acknowledges support by the Office of Naval Research under award number N00014-20-1-2725. A.A. acknowledges support by the Office of Naval Research under award number N00014-20-1-2573. P.S. acknowledges the support of the Vanier Canada Graduate Scholarship. This work made use of the NUFAB and Keck-II facilities of Northwestern University's NUANCE Center, which has received support from the SHyNE Resource (NSF ECCS-2025633), the IIN and Northwestern's MRSEC programme (NSF DMR-1720139). We thank D. Kopilovic for providing the LED spectrum and J. Gao for assisting with the SEM measurements.

Author contributions S.M.P., M.W., M.G. and E.H.S. conceived the idea and proposed the experimental and modelling design. N.L., L.A., V.C. and U.R. carried out the molecular dynamics simulation. S.M.P. fabricated all the devices and conducted the characterization. T.H., H.R.A. and K.R.G. performed XPS, UPS and IPES characterization and data analysis. W.Y. and L.X. carried out the HAADF-STEM measurements. F.T.E., M.W., S.M.Z. and M.G. conducted the photoluminescence and EQE characterization and data analysis. M.W. measured DLS. H.S. conducted cyclic voltammetry measurements and data analysis. D.C. performed UV–visible spectroscopy characterization. Y.Y. and M.G.K. measured TOF-SIMS. K.D. and A.A. performed the GIWAXS measurements. M.V., E.D.J. and D.B.K. helped with the device fabrication and material characterization. P.S. and T.F. performed the KPFM measurements. M.W., S.M.P., N.L., M.G. and E.H.S. co-wrote the manuscript. All authors contributed to data analysis, and read and commented on the manuscript.

Competing interests The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41586-023-06745-7.

Correspondence and requests for materials should be addressed to Michael Grätzel or Edward H. Sargent.

Peer review information *Nature* thanks Francis Lin and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

natureresearch

Corresponding author(s): Michael Graetzel; Edward H. Sargent

Solar Cells Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted papers reporting the characterization of photovoltaic devices and provides structure for consistency and transparency in reporting. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

For further information on Nature Research policies, including our data availability policy, see Authors & Referees.

Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	0.049 cm2
Method used to determine the device area	☒ Yes ☐ No	The active area was determined by the aperture shade mask (0.049 cm2) placed in front of the solar cell

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	Figure 4d (reverse and quasi-steady-state scans)
Voltage scan conditions *For instance: scan direction, speed, dwell times*	☒ Yes ☐ No	Supplementary Figure 21
Test environment *For instance: characterization temperature, in air or in glove box*	☒ Yes ☐ No	Supplementary Figure 21
Protocol for preconditioning of the device before its characterization	☐ Yes ☒ No	No preconditioning is needed
Stability of the J-V characteristic *Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.*	☒ Yes ☐ No	Supplementary Fig. 20b

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☐ Yes ☒ No	PCE from reverse scan matches well with those from the quasi-steady-state scan and steady-state measurements.
Related experimental data	☒ Yes ☐ No	Figure 4d and Supplementary Fig. 20

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	Figure 4c
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	The difference was found to be less than 1.5% (Main text - Solar cell efficiency and stability)
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	Not relevant

5. Calibration

Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Methods - Solar cell characterization
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	Methods - Solar cell characterization

Calculation of spectral mismatch between the reference cell and the devices under test

☒ Yes ☐ No — M = 1.019 for the certified device

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes ☐ No — 0.049 cm2

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes ☒ No — The accuracy of short-circuit current density was confirmed from EQE measurements.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes ☐ No — Newport

A copy of any certificate(s)
Provide in Supplementary Information

☒ Yes ☐ No — Supplementary Figure 21

8. Statistics

Number of solar cells tested

☒ Yes ☐ No — Figure 4b and Supplementary Figure 22

Statistical analysis of the device performance

☒ Yes ☐ No — Figure 4b and Supplementary Figure 22

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions
For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature

☒ Yes ☐ No — Figure 4f, Supplementary Figure 26 and Supplementary Figure 28